



**IILM UNIVERSITY**

School of Computer Science & Engineering

CISCO NETWORKING ACADEMY

# CISCO – AICTE Virtual Internship

Final Project Evaluation Presentation

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**PROGRAM & SEMESTER**

**B.Tech CSE • Semester 3**

**ORGANIZATION**

**Cisco VIP Internship Program**

## INTERNSHIP FOCUS

**AI**

**Applied Artificial Intelligence**

**NET**

**Cisco-style troubleshooting**

**CLI**

**Show-command diagnostics**

**AI+**

**AI-assisted root-cause analysis**

**May–August 2026**

**Cisco VIP Internship**

# From coursework to an applied AI project

The internship was structured in two phases: foundation learning followed by hands-on project work.

## PHASE 1

### Foundation learning

- Data Analytics Essentials
- Introduction to Modern AI
- Apply AI: Analyze Customer Reviews

## PHASE 2

### NetSage AI project

- AI-assisted network troubleshooting
- Python rule-based validation
- Human review + dashboard analysis

# Network symptoms rarely point to one obvious root cause

Junior engineers may know individual Cisco commands but still struggle to connect a symptom to the actual fault.

## Example symptom

**“PC Receives An IP  
But Cannot Reach a  
Server.”**

AMBIGUITY

VLAN • Gateway • DHCP • DNS • Routing •  
ACL • NAT

## What the project needed to do

- Use the symptom, topology note and show-command output as evidence.
- Suggest the likely root cause, OSI layer, next diagnostic command and fix.
- Keep a human reviewer in the loop before accepting any diagnosis.

# Build a structured, evidence-based troubleshooting workflow

The project goals were defined around data coverage, AI consistency, independent validation, review, and reporting.

**30+** **Network Troubleshooting Cases**

Dataset coverage

**8** **Fault Categories**

VLAN → Wireless

**JSON** **Structured AI Output**

Consistent comparison

**Python** **Rule-Based Checker**

Independent signal

**Human** **Review On Every Case**

Accept / Edit / Reject

**Dashboard** **Final Analysis**

Severity + agreement

# What NetSage AI covers—and what it does not

The project targets Cisco-style Packet Tracer lab troubleshooting, using provided evidence to support diagnosis.

## Fault categories

VLAN

Gateway

DHCP

DNS

Routing

ACL

NAT

Wireless

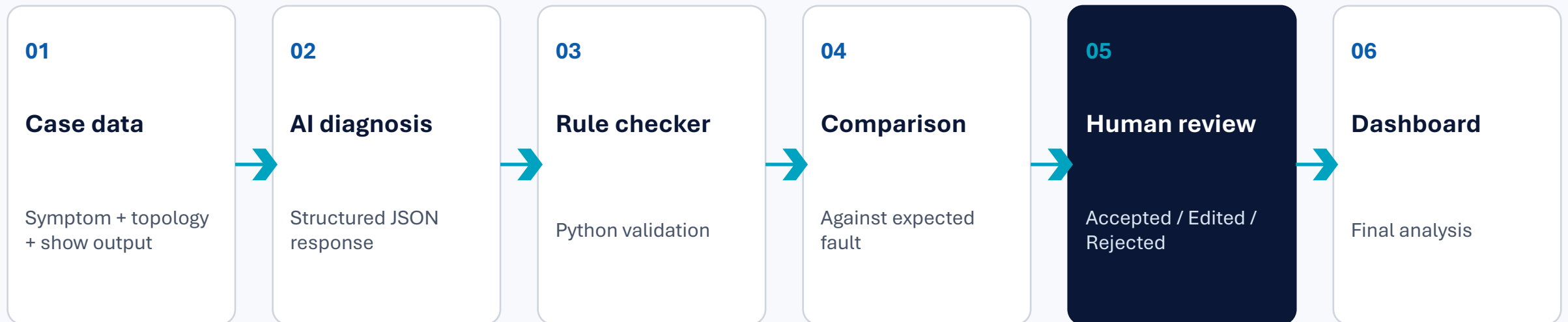
## Boundary

### Decision support —not autonomous change

The AI proposes.  
The rule checker cross-checks.  
The human reviewer decides.

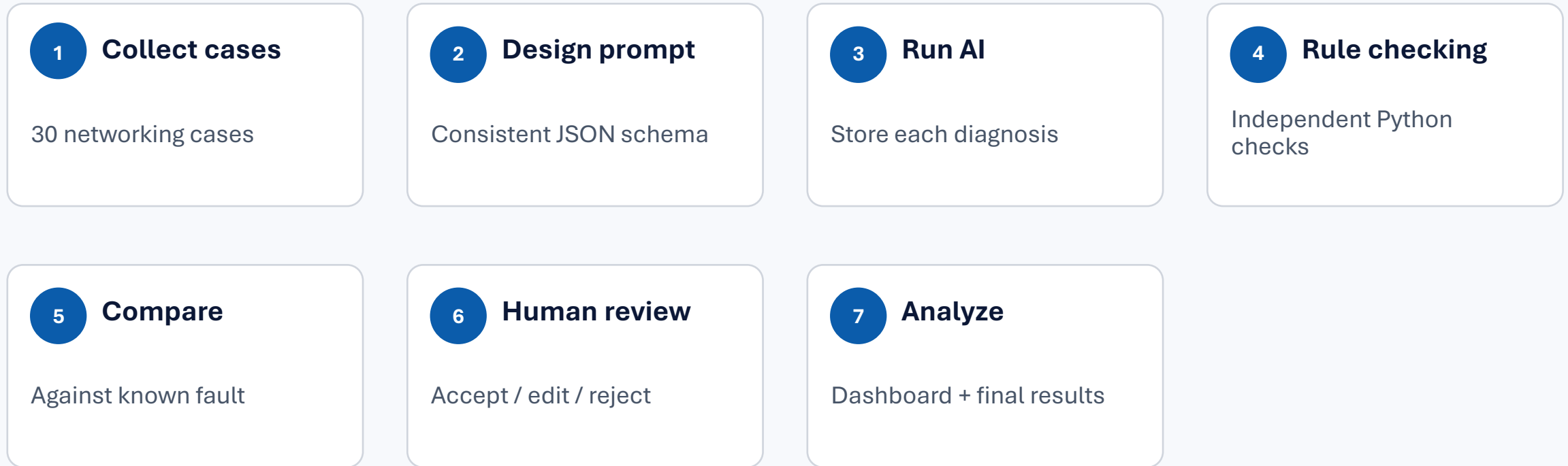
# A sequential pipeline with multiple checks

Each troubleshooting case passes through AI diagnosis, deterministic validation, comparison and human review before final reporting.



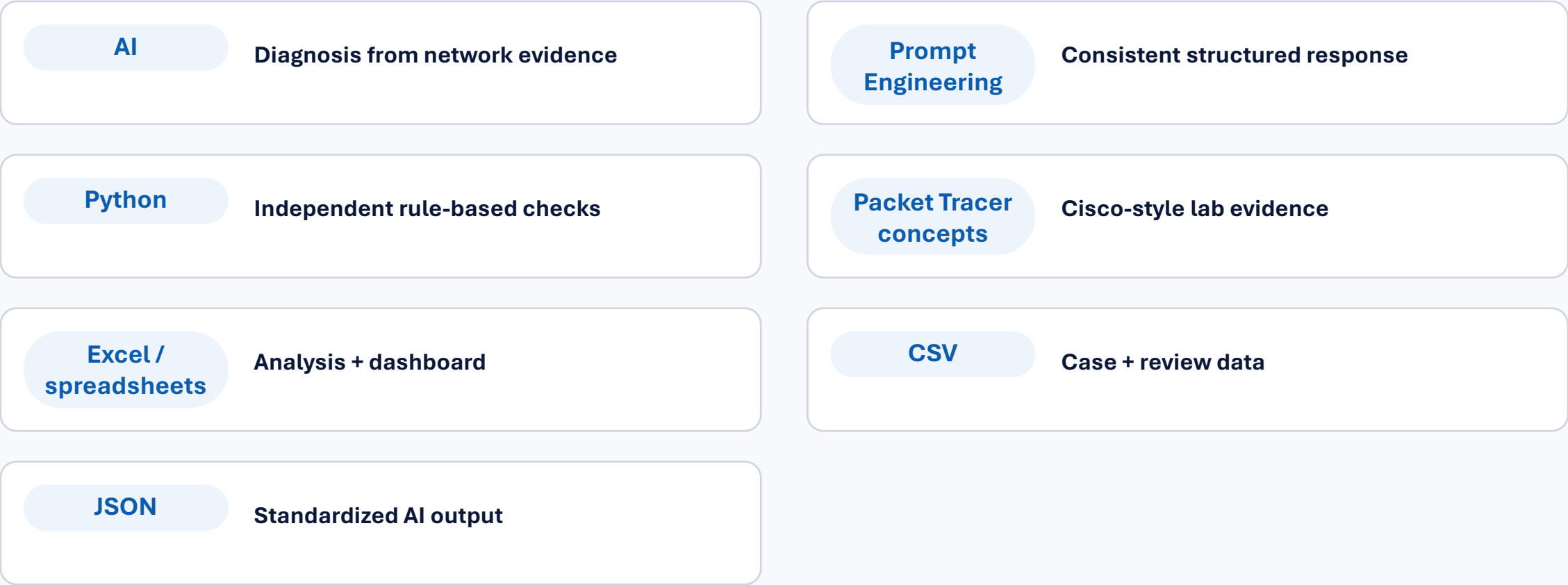
# Seven steps from case collection to final analysis

The process was designed to make AI outputs comparable, reviewable and auditable.



# Tools were combined around one practical workflow

The report lists AI, prompt engineering, Python, Packet Tracer concepts, spreadsheets, CSV and JSON as the main implementation pieces.





# Every response is structured for review and comparison

A fixed JSON-style schema was used so outputs could be evaluated consistently across cases.

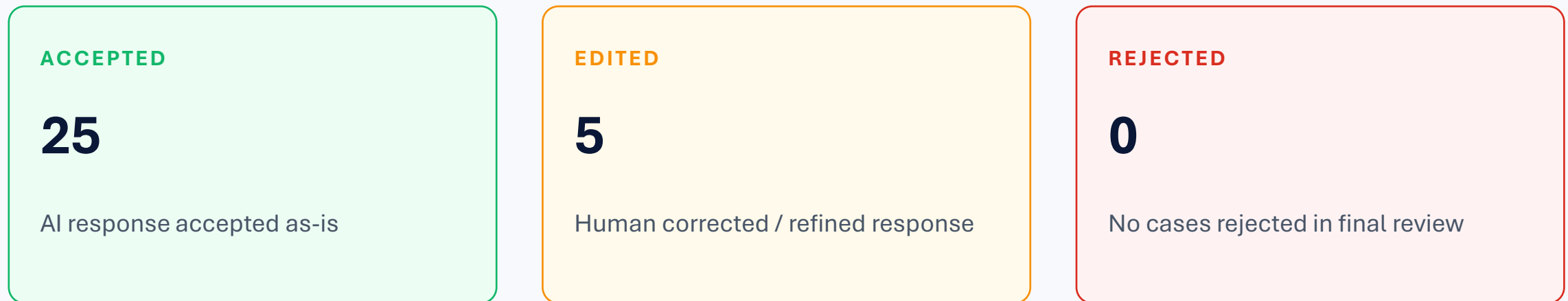
```
{  "root_cause": "...",  
  "osi_layer": "...",  
  "confidence": "...",  
  "evidence": ["..."],  
  "next_command": "...",  
  "fix_steps": ["..."]  
}
```

## Why this structure matters

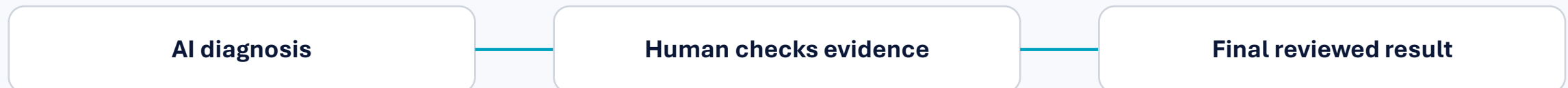
- Separates diagnosis from evidence.
- Makes confidence explicit instead of implicit.
- Creates a consistent object for comparison and review.
- Supports downstream spreadsheet/dashboard analysis.

# Human review is part of the system—not an afterthought

Every AI diagnosis is reviewed before it is considered trustworthy. Corrections are recorded in the human-review log.

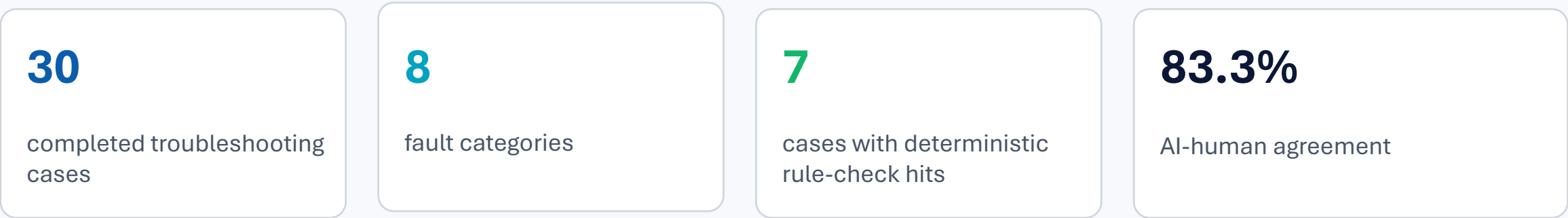


## Review decision flow



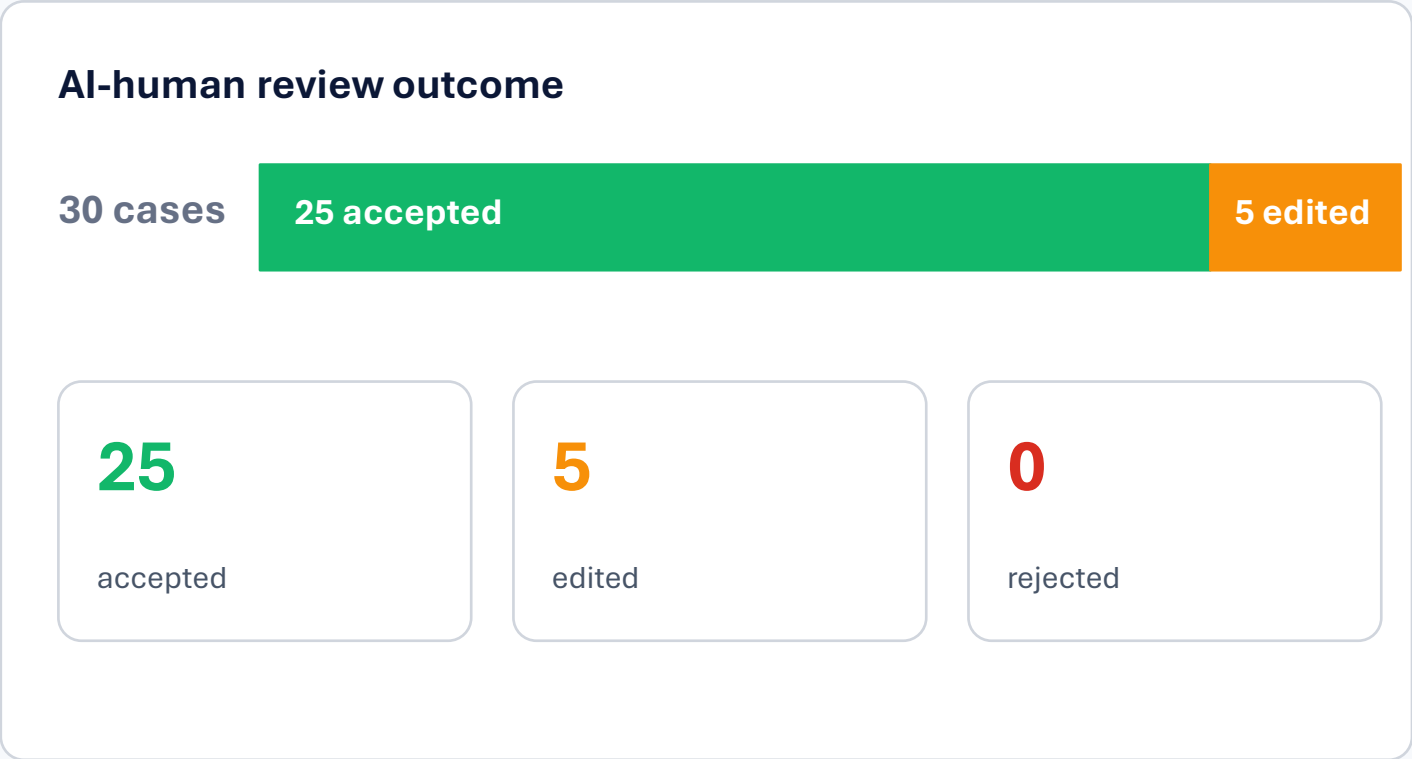
# 30 cases across 8 troubleshooting categories

Each case includes the symptom, topology note, show-command output, expected fault, OSI layer, concept and severity.



# The strongest evidence is the reviewed outcome—not raw AI output

The final review produced 25 accepted and 5 edited responses, with no rejected cases.



**83.3%**

**AI-human agreement**

**7 / 30**

cases with a deterministic issue detected by the rule checker

# What the internship demonstrated

The project connected AI concepts, data handling, Python validation and network troubleshooting into one applied workflow.

## Project outcomes

- Structured troubleshooting support instead of isolated command lookup.
- Evidence-based responses grounded in the supplied case data.
- Independent deterministic checks to complement AI output.
- Documented human corrections for responsible AI use.

## Skills gained

Artificial Intelligence

Prompt design

Python

Networking troubleshooting

Data organization

Responsible AI

## Overall result

**A human-supervised decision-support workflow for lab networking.**

# NetSage AI turns troubleshooting into a reviewable workflow

The project's value is not “AI replaces the engineer”; it is that AI, deterministic checks and human judgment work together.

**AI**

Generate structured diagnosis

**RULES**

Cross-check mechanical faults

**HUMAN**

Approve or correct

**Final takeaway**

**Evidence first. AI assists. Humans remain accountable.**

# THANK YOU

**Aditya Raj**

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